

Samuel Fred

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Summary

Data Engineer with 8 years of experience in designing, developing, and optimizing scalable data pipelines, architectures, and distributed data processing systems. Expertise in ETL/ELT processes, data modeling, and data warehousing solutions using technologies like Snowflake, Redshift, BigQuery, and Databricks. Proficient in big data technologies such as Apache Spark, Hadoop, Kafka, and Flink, with hands-on experience in real-time and batch data processing. Skilled in cloud-based data platforms (AWS, Azure, GCP) and infrastructure-as-code (Terraform, CloudFormation). Strong programming skills in Python, SQL, and Scala, with experience in workflow orchestration tools like Airflow, dbt, and Prefect. Adept at collaborating with cross-functional teams to implement data governance, optimize query performance, and enable data-driven decision-making.

Skills

Programming & Scripting: Python, SQL (PostgreSQL, MySQL, SQL Server), Java, Scala, Shell Scripting

Big Data & Distributed Computing: Apache Spark (PySpark, Spark SQL), Hadoop (HDFS, MapReduce, YARN), Apache Hive, Apache Flink, Apache Kafka, Dask

Databases & Data Storage: PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Cassandra, DynamoDB, Snowflake, Redshift, BigQuery, AWS S3, Azure Data Lake, Google Cloud Storage

ETL & Data Pipelines: Apache Airflow, dbt (Data Build Tool), Talend, Informatica, AWS Glue, Apache NiFi

Cloud Platforms & Services: AWS (S3, Lambda, Redshift, Glue, EMR, Athena, Kinesis), Azure (Data Factory, Synapse Analytics, Databricks, Cosmos DB), GCP (BigQuery, Dataflow, Cloud Composer, Pub/Sub)

Business Intelligence & Data Visualization: Tableau, Power BI, Looker, Superset, Metabase

DevOps & CI/CD: Docker, Kubernetes, Terraform, Git (GitHub, GitLab), Jenkins, CI/CD Pipelines

Data Governance & Security: Data Cataloging & Lineage (Apache Atlas, Alation), Data Quality (Great Expectations), GDPR, HIPAA Compliance, Role-Based Access Control (RBAC), Encryption & Security Policies

Experience

Lead Data Engineer

DataSap

03/2022 - Present

- Designed and optimized distributed data pipelines with Apache Spark, Kafka, and Airflow, cutting processing time by 50%.
- Architected and managed a scalable cloud-based data platform on AWS (S3, Redshift, Glue, Lambda, EMR), enhancing performance and reliability.
- Established and enforced **data governance and security policies** using Apache Ranger, AWS IAM, and Lake Formation.
- Led a team of 5+ **data engineers**, developing CI/CD pipelines for data workflows using **Terraform, GitHub Actions, and Docker**.
- Developed **real-time data streaming solutions** with Apache Kafka, AWS Kinesis, and Apache Flink, improving event-driven processing.
- Integrated **monitoring and logging** with Prometheus, Grafana, and AWS CloudWatch, boosting system reliability.

- Implemented **cost-optimization strategies** for data storage and computing, reducing cloud expenses by **30%**.

Senior Data Engineer

Roche

05/2020 - 02/2022

- Led the migration of **on-premises databases** to **AWS Redshift** and **Google BigQuery**, enhancing performance and cutting storage costs by **25%**.
- Designed and deployed a **data lakehouse architecture** leveraging **Delta Lake**, **Apache Iceberg**, and **AWS Glue**, streamlining ETL workflows.
- Developed **real-time streaming pipelines** using **Kafka**, **Kinesis**, and **Apache Flink**, enabling low-latency event processing.
- Optimized **SQL queries and indexing strategies** in **Snowflake**, **PostgreSQL**, and **MySQL**, reducing query execution time by **60%**.
- Automated **data transformation and modeling** with **dbt (Data Build Tool)**, ensuring consistency and efficiency.
- Implemented **data cataloging solutions** with **AWS Glue Data Catalog** and **Apache Atlas** for improved metadata management.
- Established **role-based access controls (RBAC)** using **AWS IAM**, **Snowflake RBAC**, and **Databricks Unity Catalog**, enhancing data security.

Associate Data Engineer

AB InBev

03/2018 - 04/2020

- Designed and managed **ETL pipelines** using **Apache Airflow**, **Python**, and **SQL**, enhancing data ingestion efficiency by **40%**.
- Built and optimized **batch processing workflows** with **Apache Spark (PySpark)** and **Hadoop**, significantly improving data processing speed.
- Developed **data warehouse schemas** in **Snowflake**, **Redshift**, and **BigQuery**, enabling faster analytics and query performance.
- Created and maintained **API-based data ingestion pipelines**, integrating with **RESTful APIs**, **Kafka**, and **SFTP sources**.
- Automated **data quality checks and anomaly detection** using **Great Expectations** and **AWS Lambda**, ensuring high data accuracy.
- Developed **containerized ETL jobs** with **Docker** and **Kubernetes**, improving scalability and deployment efficiency.

Data Analyst

AlphaBridge

05/2015 - 02/2018

- Analyzed **large datasets** using **SQL**, **Python**, and **Excel** to uncover trends and insights that enhanced business decision-making.
- Designed **interactive dashboards and reports** in **Tableau** and **Power BI**, improving data visibility for stakeholders.
- Developed **predictive models** leveraging **machine learning techniques** to optimize marketing campaigns, boosting ROI by **15%**.
- Automated **data collection and cleaning processes**, reducing manual effort by **40%** while improving data accuracy.
- Conducted **A/B testing and statistical analysis** to evaluate the effectiveness of business strategies.
- Partnered with **cross-functional teams** to translate complex data insights into actionable business recommendations.
- Implemented a **Grafana dashboard** for real-time data monitoring and insights.

Projects

1. Real-Time Fraud Detection System

- Designed a **real-time fraud detection system** using **Apache Flink & Kafka**, processing **1M+ transactions/sec**.
- Reduced fraudulent activities by **30%** using machine learning integration.

2. Scalable Data Lake & ETL Framework

- Built a **centralized data lake** on AWS S3, integrating with **Glue & Redshift Spectrum** for efficient querying.
- Developed an **ETL automation pipeline** that processes **billions of rows daily**.

3. Event-Driven Data Pipeline for IoT Analytics

- Created an **IoT data pipeline** using **AWS Kinesis, Lambda & DynamoDB** for real-time device monitoring.
- Enabled predictive maintenance for **industrial sensors**, reducing downtime by **20%**.

4. End-to-End Cloud Data Warehouse Migration

- Led the migration of **on-premise Oracle DWH to Google BigQuery**, reducing operational costs by **35%**.
- Improved query performance by **2x** using partitioning & clustering techniques.

5. Automated Reporting & Dashboards

- Developed **Power BI dashboards** that visualize sales and marketing data for C-level executives.
- Automated **ETL workflows** using **dbt & Airflow**, saving 50+ engineering hours per month.

Education

Bachelor's Degree